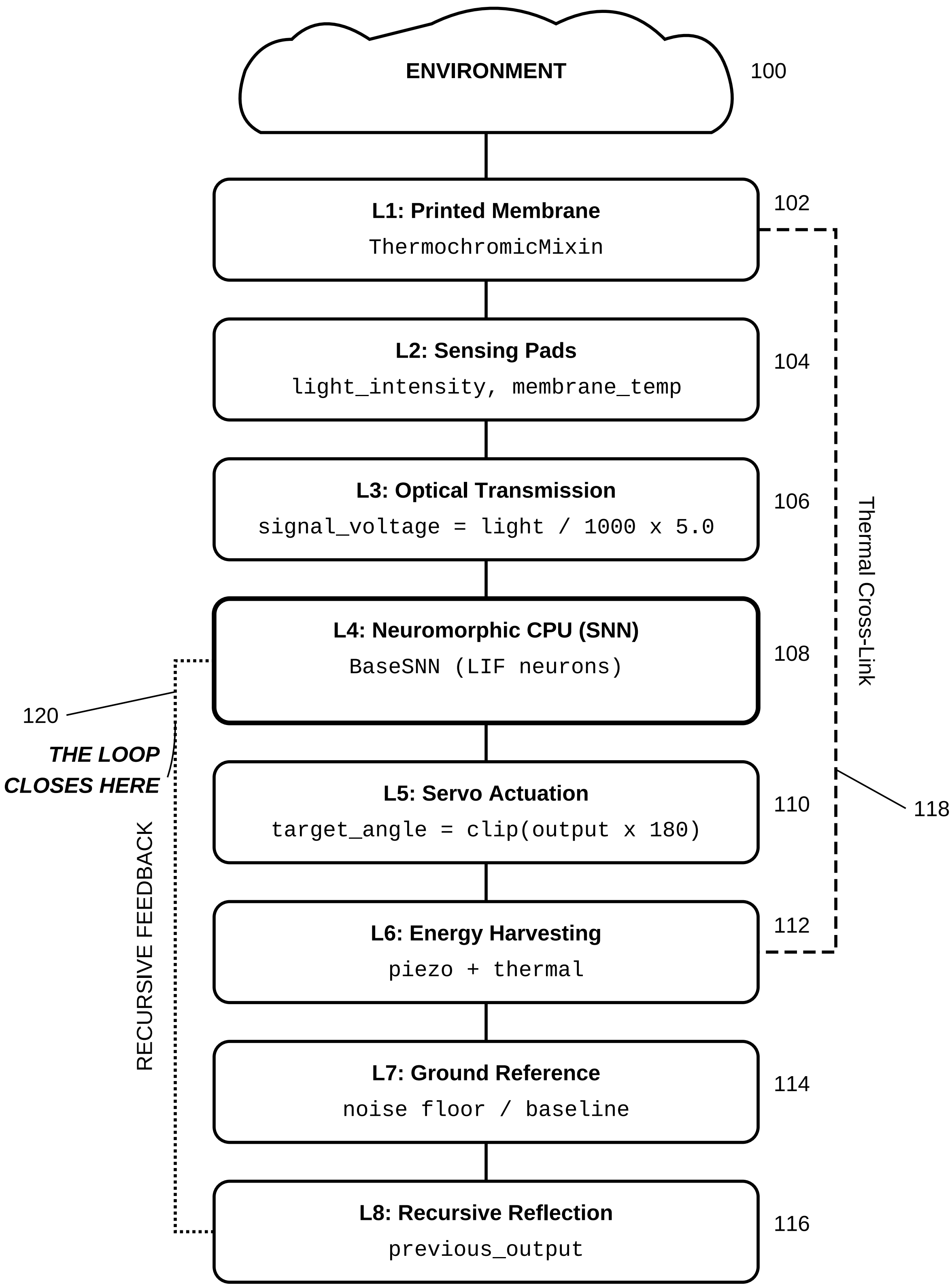
PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems

Drawing Sheets: 8 figures

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026



ENVIRONMENT
100
L1: Printed Membrane
ThermochromicMixin
102
L2: Sensing Pads
light_intensity, membrane_temp
104
L3: Optical Transmission
signal_voltage = light / 1000 x 5.0
106
Thermal Cross-Link
L4: Neuromorphic CPU (SNN)
BaseSNN (LIF neurons)
108
120
THE LOOP
CLOSES HERE
L5: Servo Actuation
target_angle = clip(output x 180)
110
118
RECURSIVE FEEDBACK
L6: Energy Harvesting
piezo + thermal
112
L7: Ground Reference
noise floor / baseline
114
L8: Recursive Reflection
previous_output
116

FIG. 1

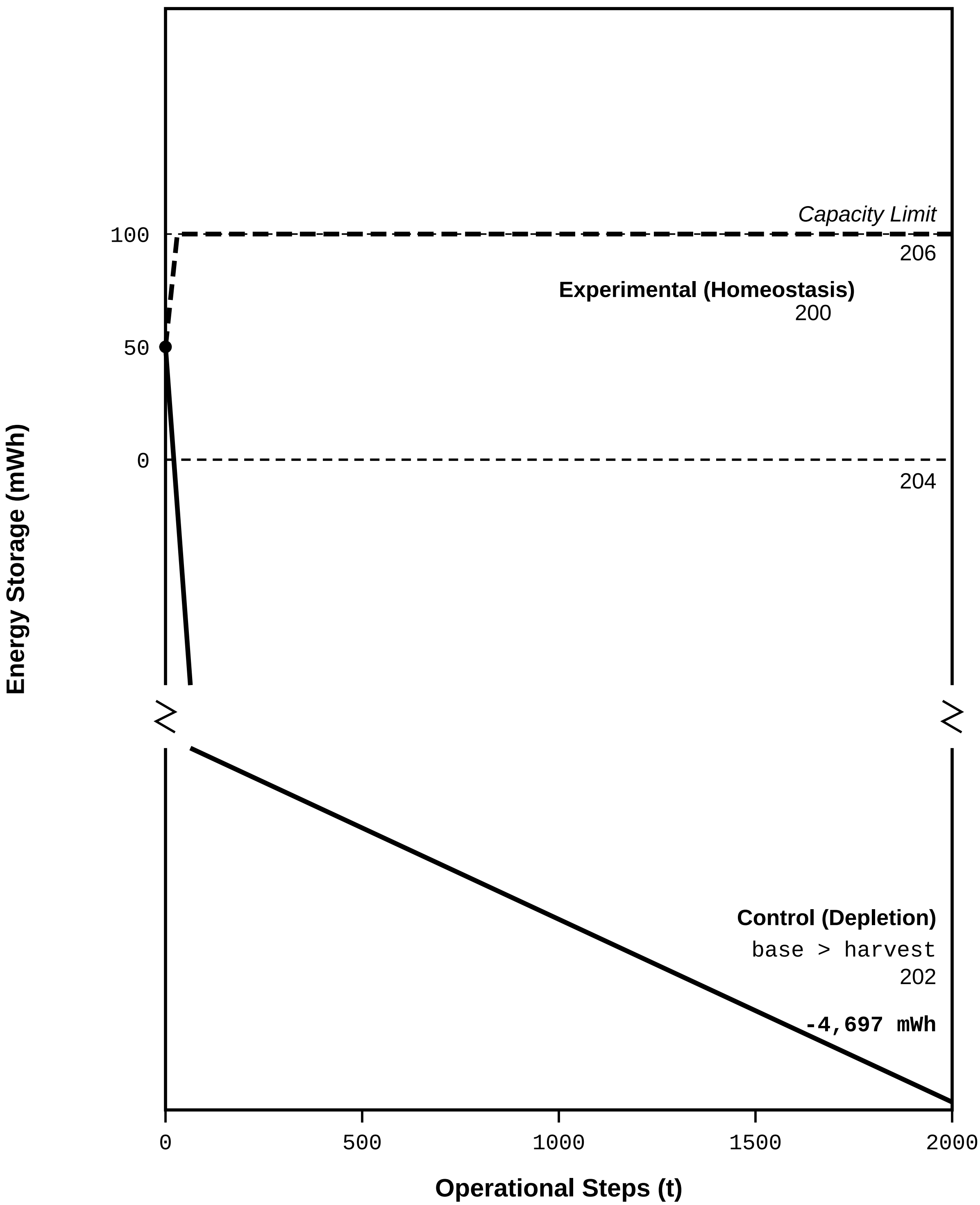

Capacity Limit
100
206
Experimental (Homeostasis)
200
50
0
204
Energy Storage (mWh)
Control (Depletion)
base > harvest
202
-4,697 mWh
0
500
1000
1500
2000
Operational Steps (t)

FIG. 2

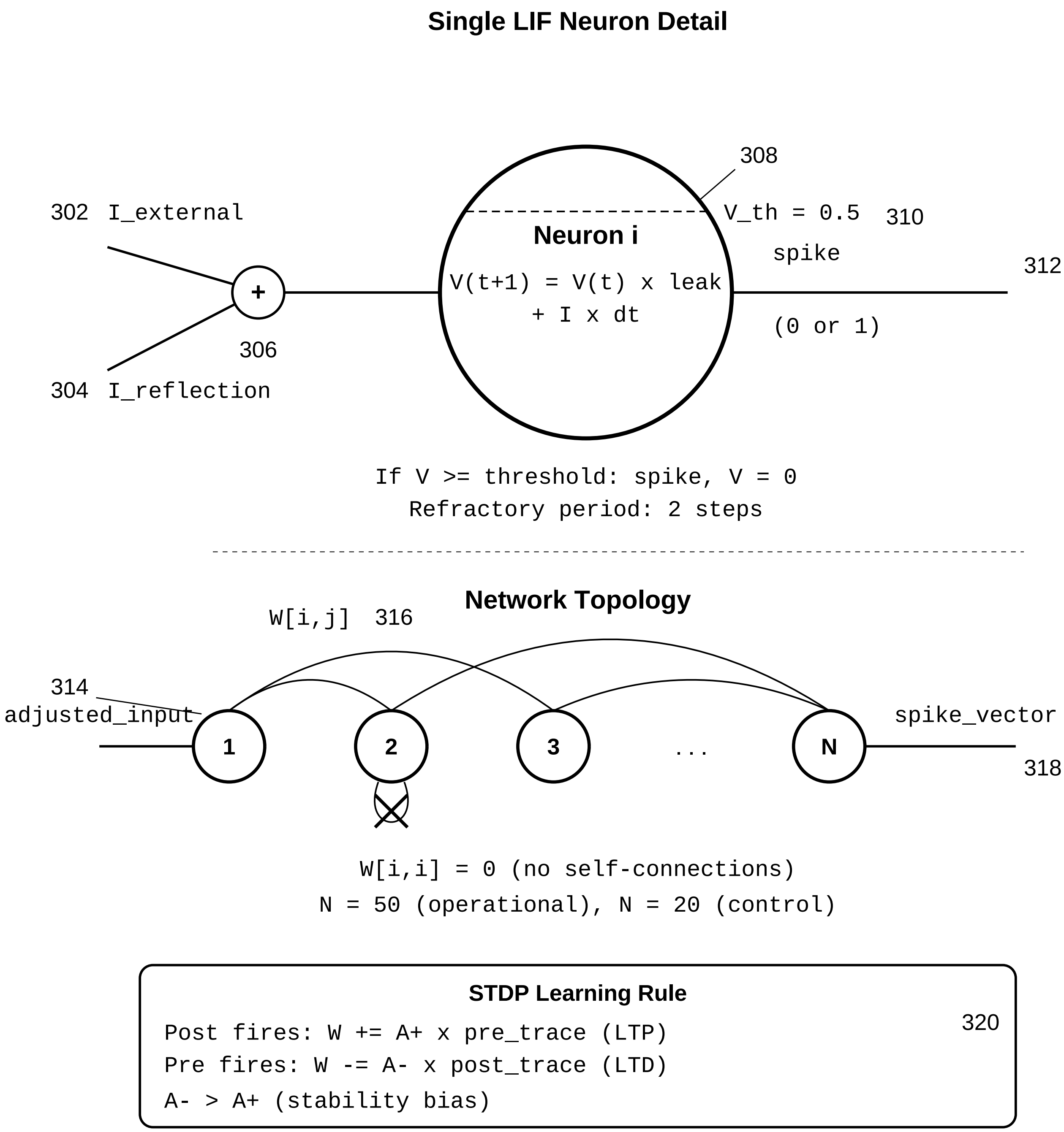
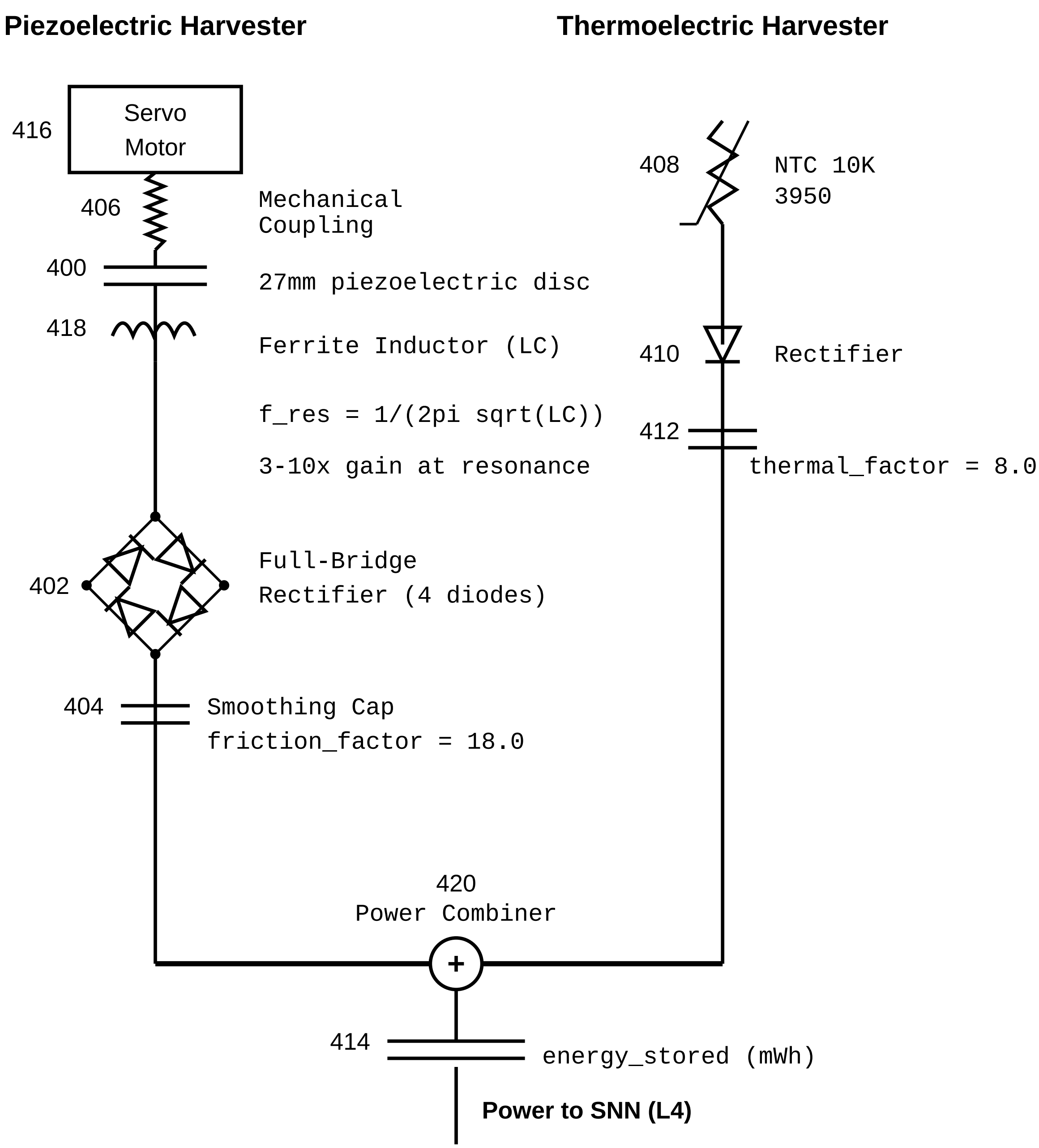

Single LIF Neuron Detail
302 I_external
304 I_reflection
+
306
Neuron i
V(t+1) = V(t) x leak
+ I x dt
308
V_th = 0.5
310
spike
(0 or 1)
312
If V >= threshold: spike, V = 0
Refractory period: 2 steps
Network Topology
W[i,j] 316
314
adjusted_input
1
2
3
...
N
spike_vector
318
W[i,i] = 0 (no self-connections)
N = 50 (operational), N = 20 (control)
STDP Learning Rule
320
Post fires: W += A+ x pre_trace (LTP)
Pre fires: W -= A- x post_trace (LTD)
A- > A+ (stability bias)

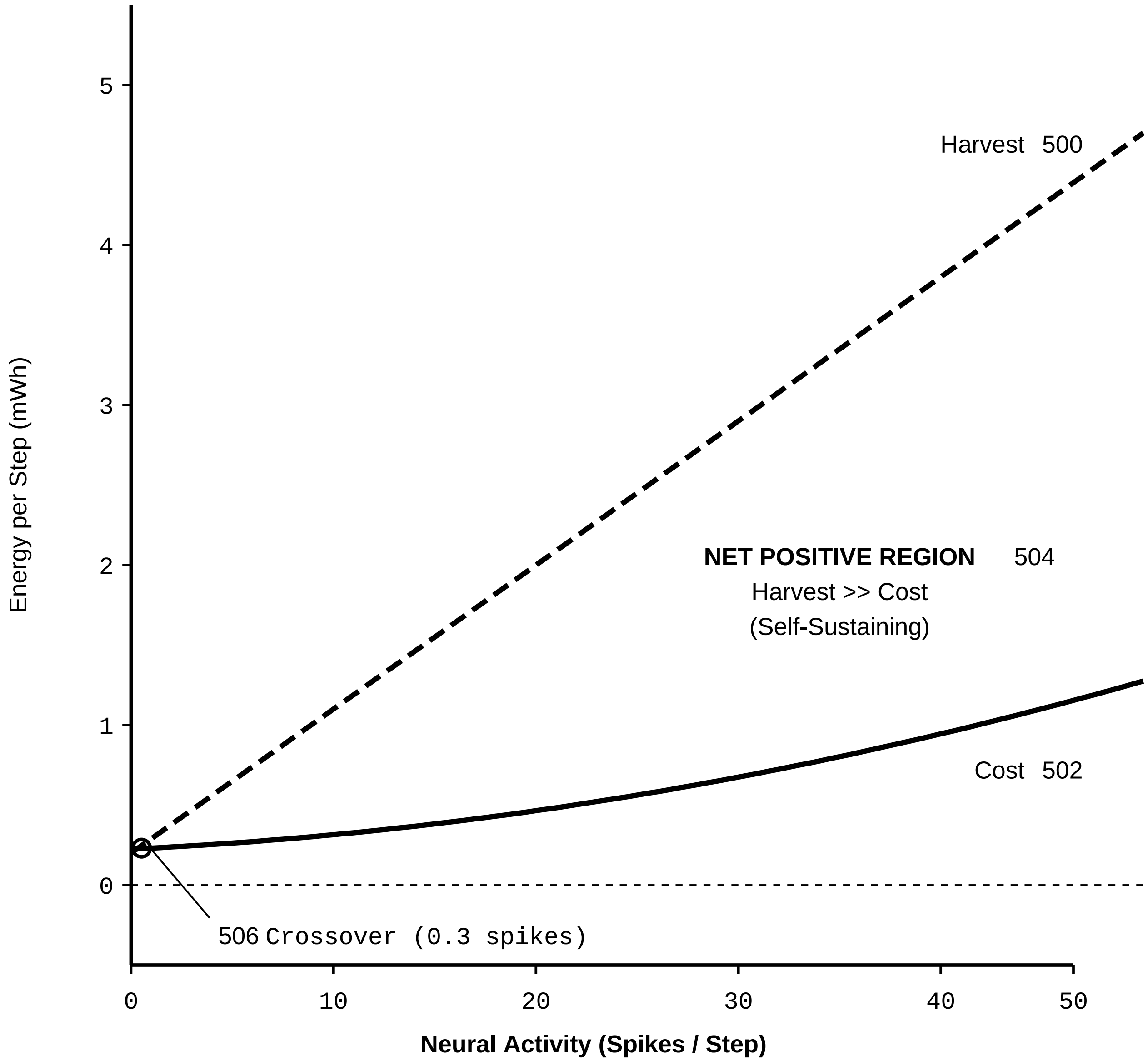
FIG. 3



Power Budget

Harvest = friction_harvest + thermal_harvest
Cost = base_consumption + activity_cost x spike_count^2
Net = Harvest - Cost

FIG. 4

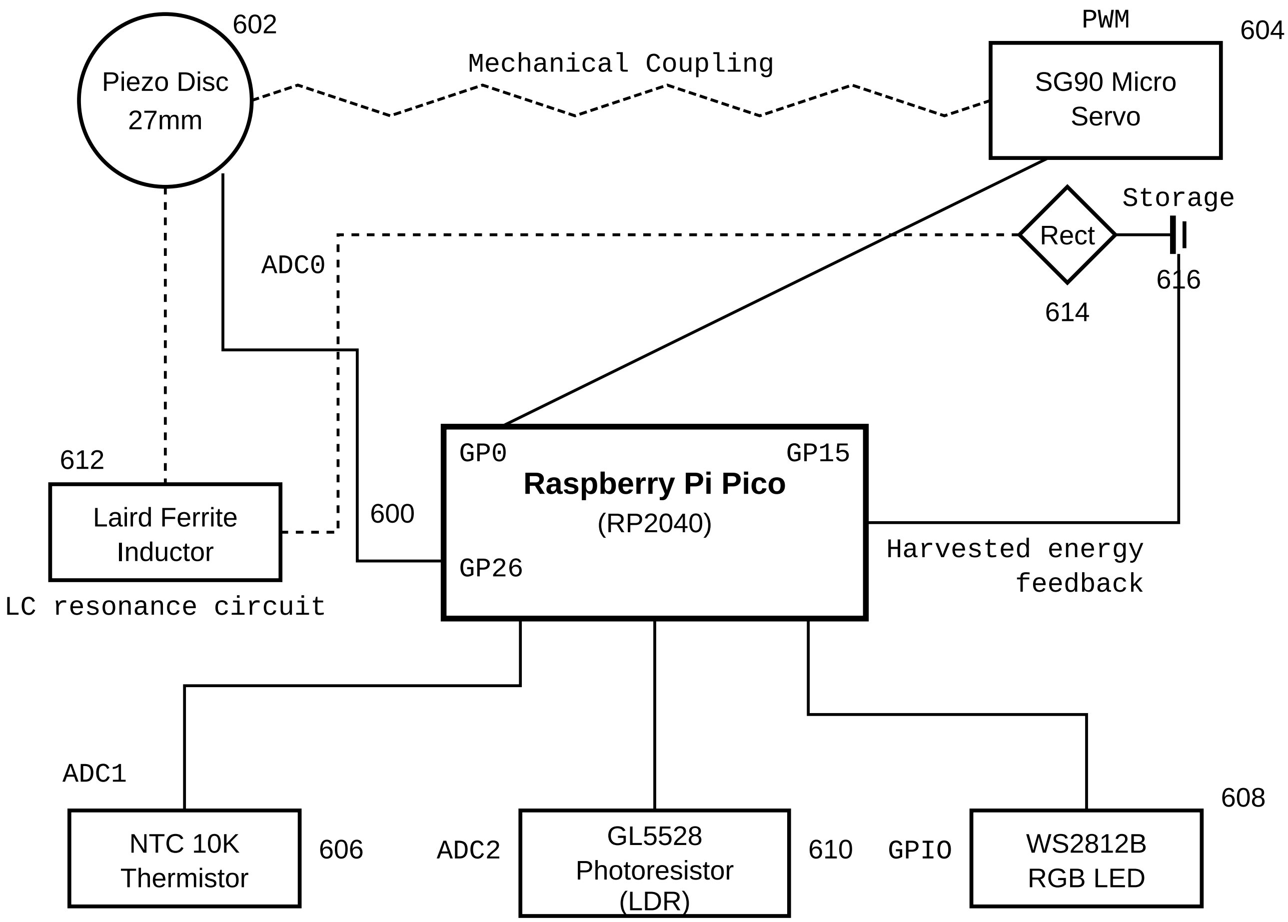



```
Harvest = 0.20 + (0.09 x spikes) mWh/step              508
Cost = 0.225 + (0.006 x s) + (0.0003 x s^2) mWh/step
Net = Harvest - Cost
```

Zone 1: Quiescent Drain

0 - 1 Spikes: Cost > Harvest

FIG. 5



Component Summary

- Raspberry Pi Pico (RP2040)
- SG90 Micro Servo
- 27mm Piezoelectric Disc
- NTC 10K Thermistor
- Photoresistor (LDR)
- WS2812B RGB LED
- Misc (wires, board, inductor)

Signal / Control

Energy Harvesting

FIG. 6

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VALIDATION RESULTS SUMMARY

Claim	Criterion	Measured	Threshold	Result
Phase 7 Control Baseline (EnergyConfig)				
A	Closed-loop continuity	2000 pts	2000	**PASS**
B	Boundedness	bounded	bounded	**PASS**
C	Robustness (noise)	Std(theta) <= 45	<= 45	**PASS**
D	Input-output gain	corr >= 0.2	>= 0.2	**PASS**
E	Saturation ratio	sat <= 0.20	<= 0.20	**PASS**
Phase 7 Full Integration (Control confirms energy depletion)				
--	Energy at 2000 steps	-4,697 mWh	< 0	**CONFIRMED**
MCP Integration (BalancedEnergyConfig)				
--	Energy at 2000 steps	100.0 mWh (capped)	> 0	**CONFIRMED**
Phase 8 STDP				
F8.1	Weight entropy decrease	2.95 to 2.02 bits	decrease	**PASS**
F8.2	STDP MI > 1.2x frozen	6.52x	> 1.20	**PASS**
F8.3	Weight convergence	0.00%	< 10%	**PASS**

STDP Learning Efficacy (Claim F8.2)

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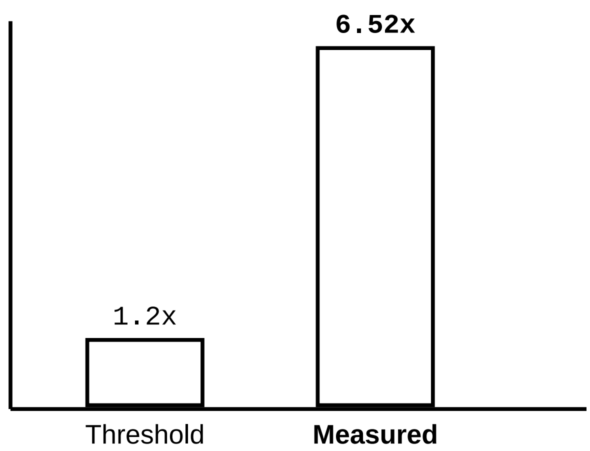
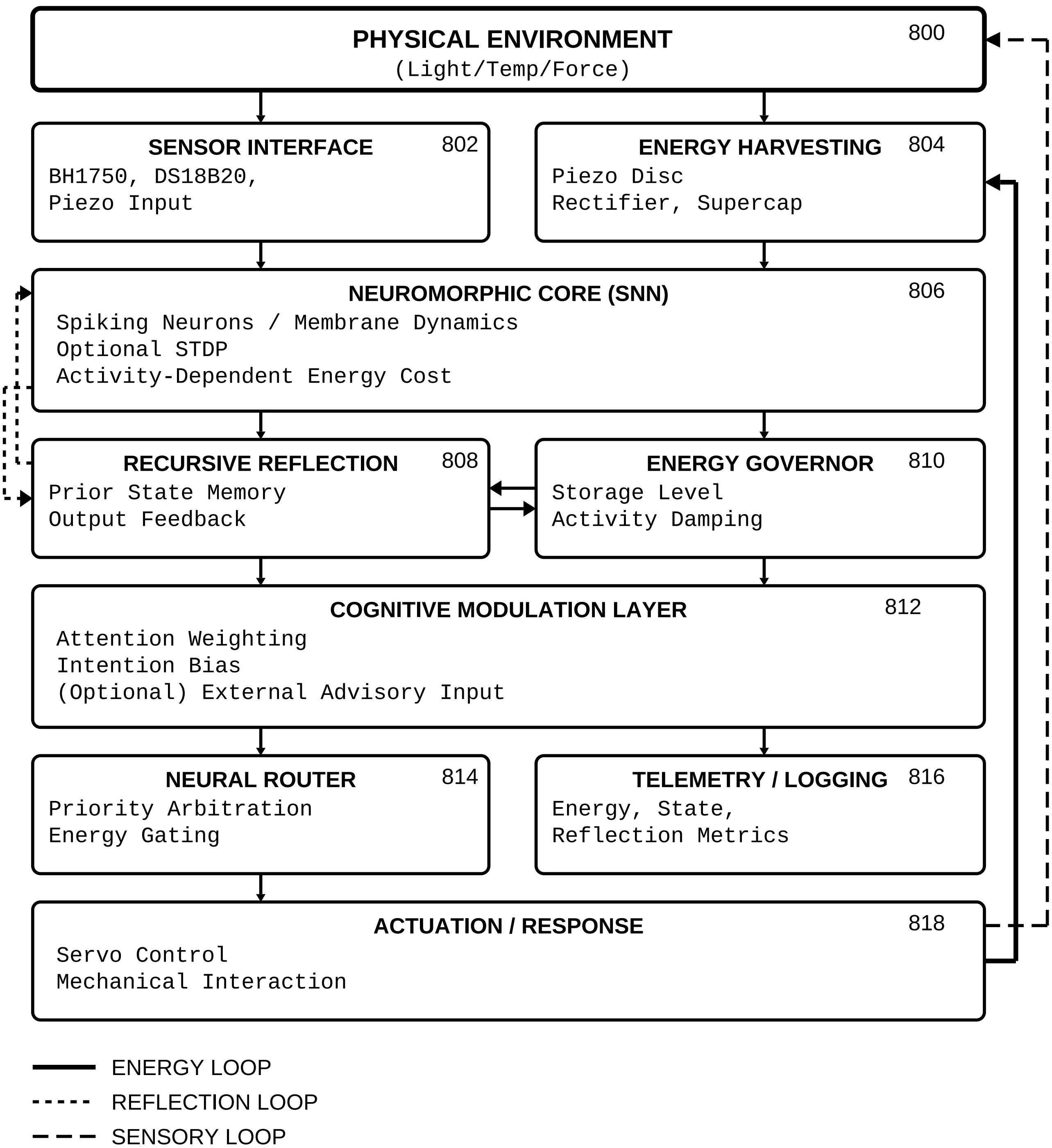


All claims validated. System is reproducible via git tags.

FIG. 7



All computation is constrained by real-time energy availability derived from mechanical interaction.

FIG. 8

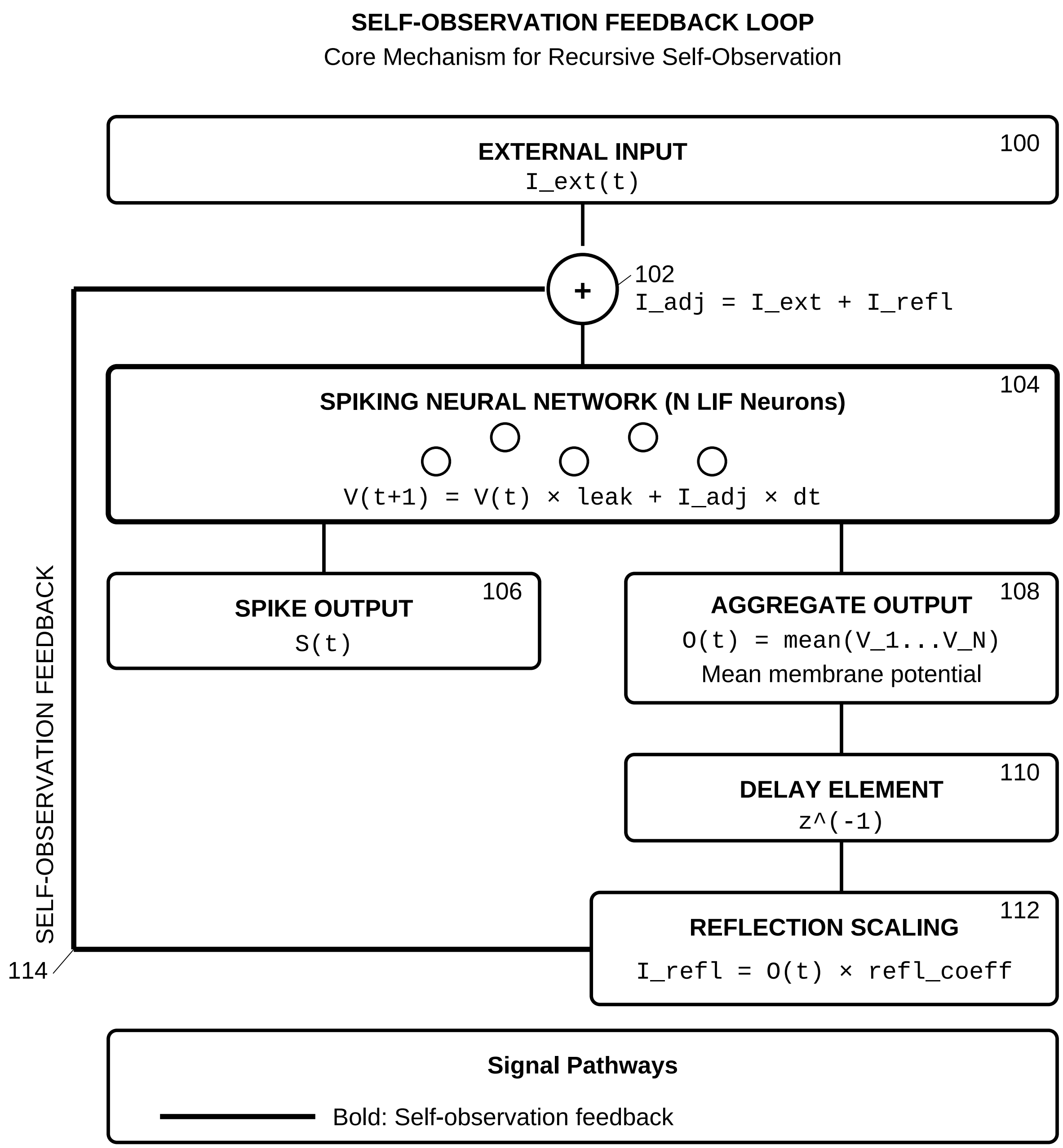
PATENT B

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

Drawing Sheets: 6 figures

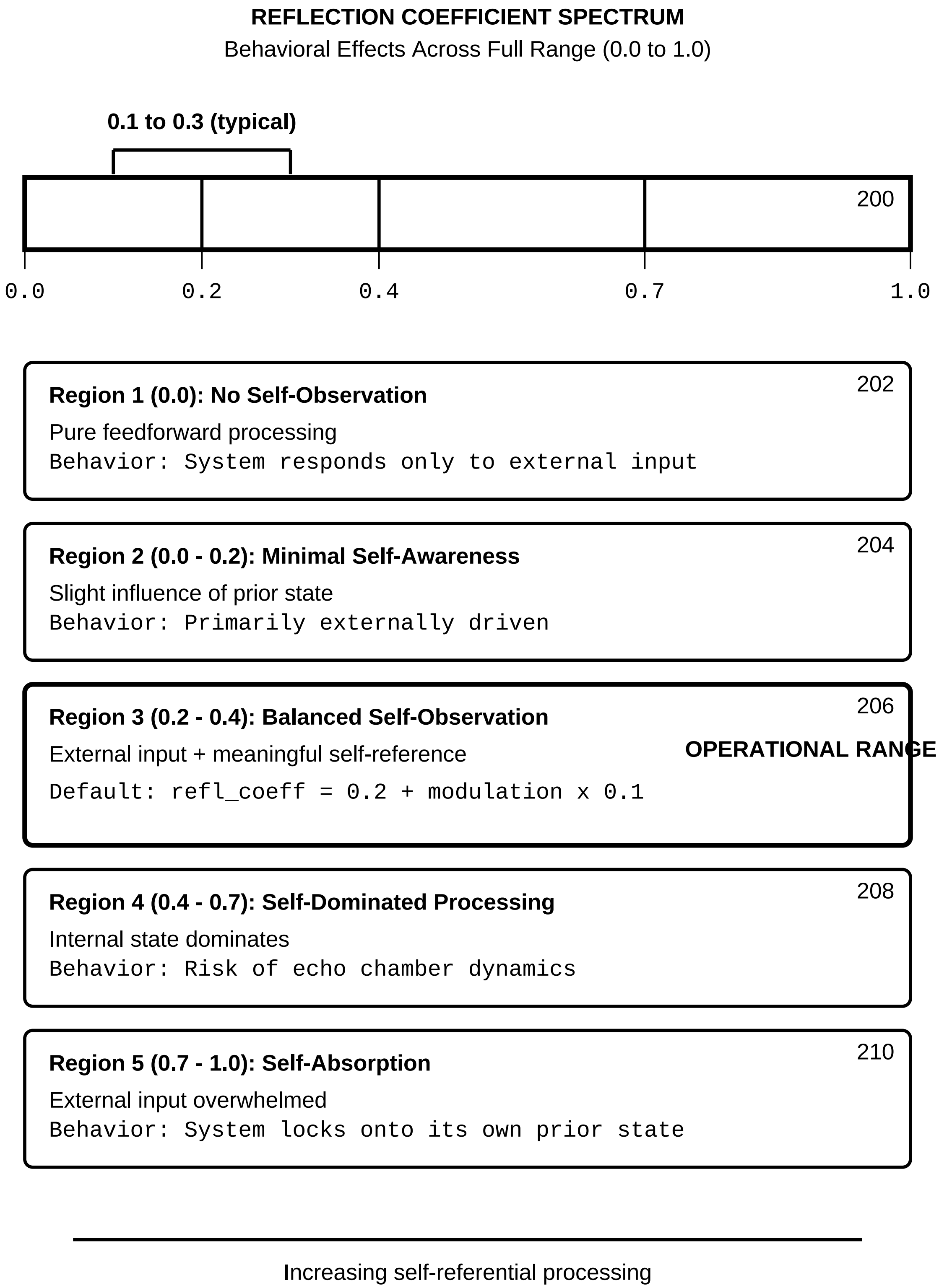
Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026



Self-observation: network observes its own prior state.
Reflection coefficient: 0.0-1.0. Operational: 0.1-0.3.

FIG. 1



The reflection coefficient is the system's self-awareness dial

FIG. 2

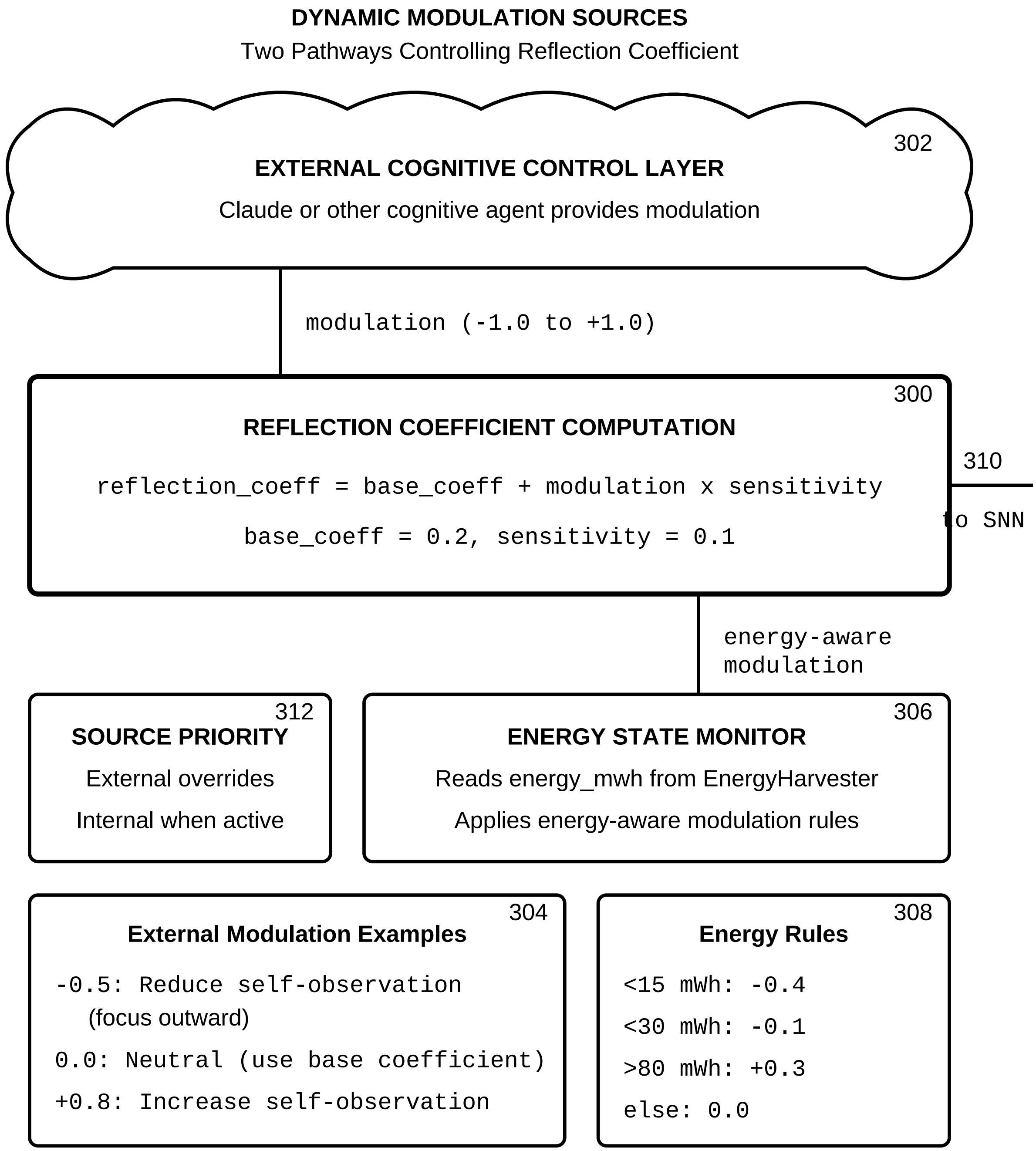

DYNAMIC MODULATION SOURCES
Two Pathways Controlling Reflection Coefficient
302
EXTERNAL COGNITIVE CONTROL LAYER
Claude or other cognitive agent provides modulation
modulation (-1.0 to +1.0)
300
REFLECTION COEFFICIENT COMPUTATION
reflection_coeff = base_coeff + modulation x sensitivity
base_coeff = 0.2, sensitivity = 0.1
310
to SNN
energy-aware
modulation
312
SOURCE PRIORITY
External overrides
Internal when active
306
ENERGY STATE MONITOR
Reads energy_mwh from EnergyHarvester
Applies energy-aware modulation rules
304
External Modulation Examples
-0.5: Reduce self-observation
(focus outward)
0.0: Neutral (use base coefficient)
+0.8: Increase self-observation
308
Energy Rules
<15 mWh: -0.4
<30 mWh: -0.1
>80 mWh: +0.3
else: 0.0

FIG. 3

SELF-REFERENTIAL LEARNING LOOP

(STDP + Self-Observation)

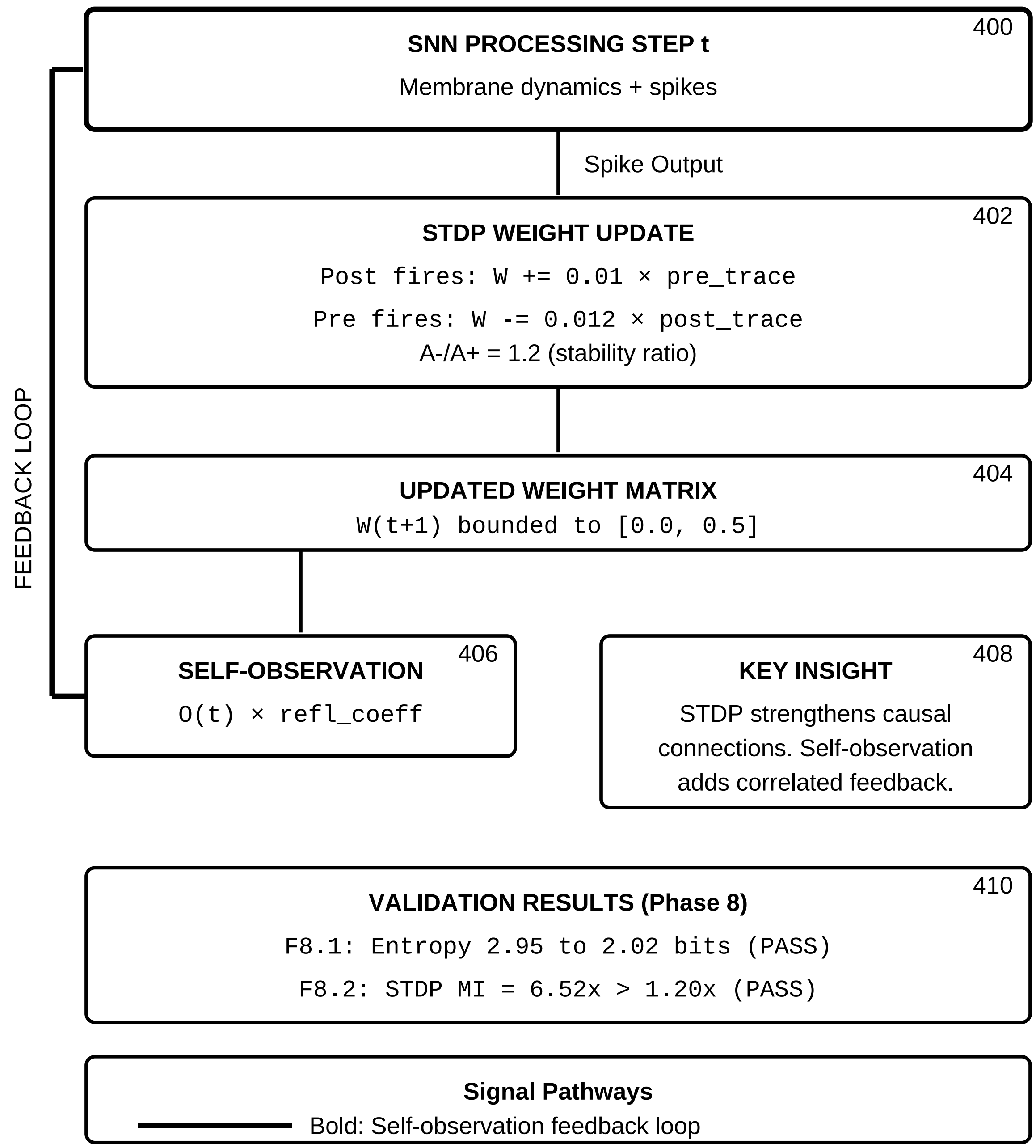
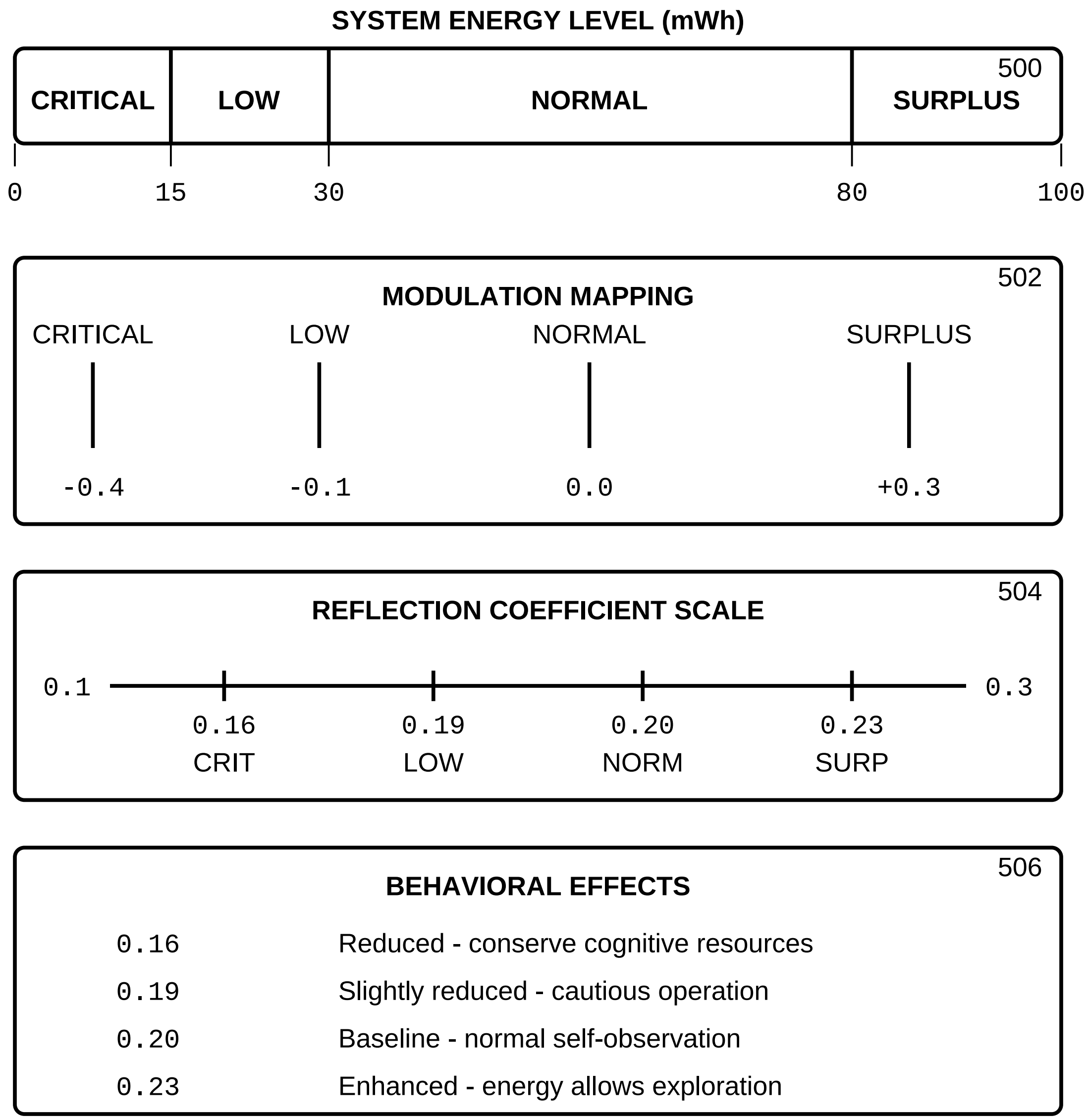


FIG. 4

ENERGY-AWARE SELF-OBSERVATION REGULATION

Reflection Coefficient Modulation Based on System Energy State

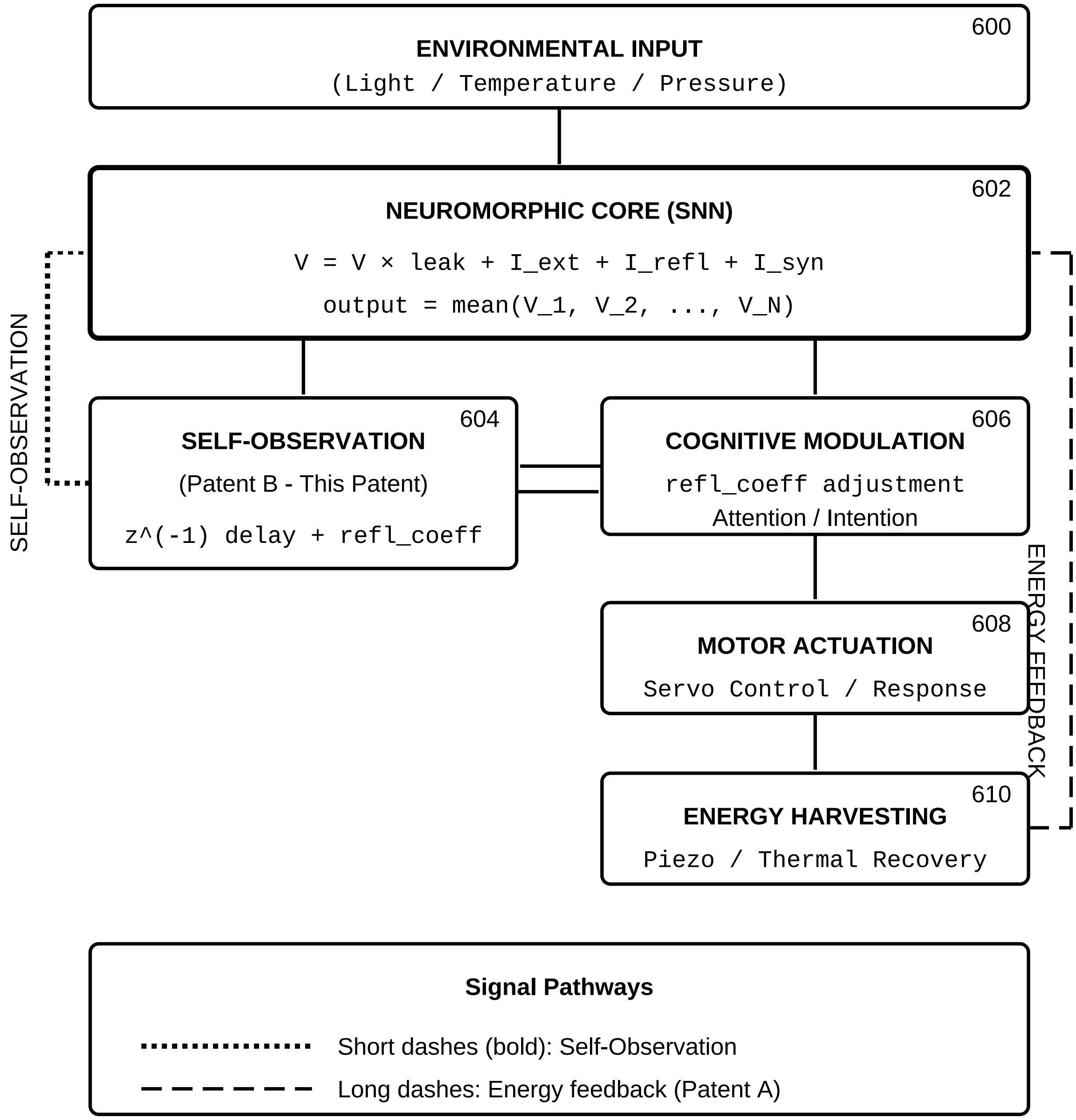


Energy-aware regulation ensures efficient cognitive resource management.
Low energy reduces self-observation; surplus energy enables exploration.

FIG. 5

END-TO-END SIGNAL FLOW

Self-Observation Integration in Complete System



The self-observation pathway feeds the network's own aggregate state back as explicit input at configurable gain (0.0 to 1.0).

FIG. 6

PATENT C

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

Drawing Sheets: 7 figures

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026

SYSTEM ARCHITECTURE WITH COGNITIVE FALLBACK

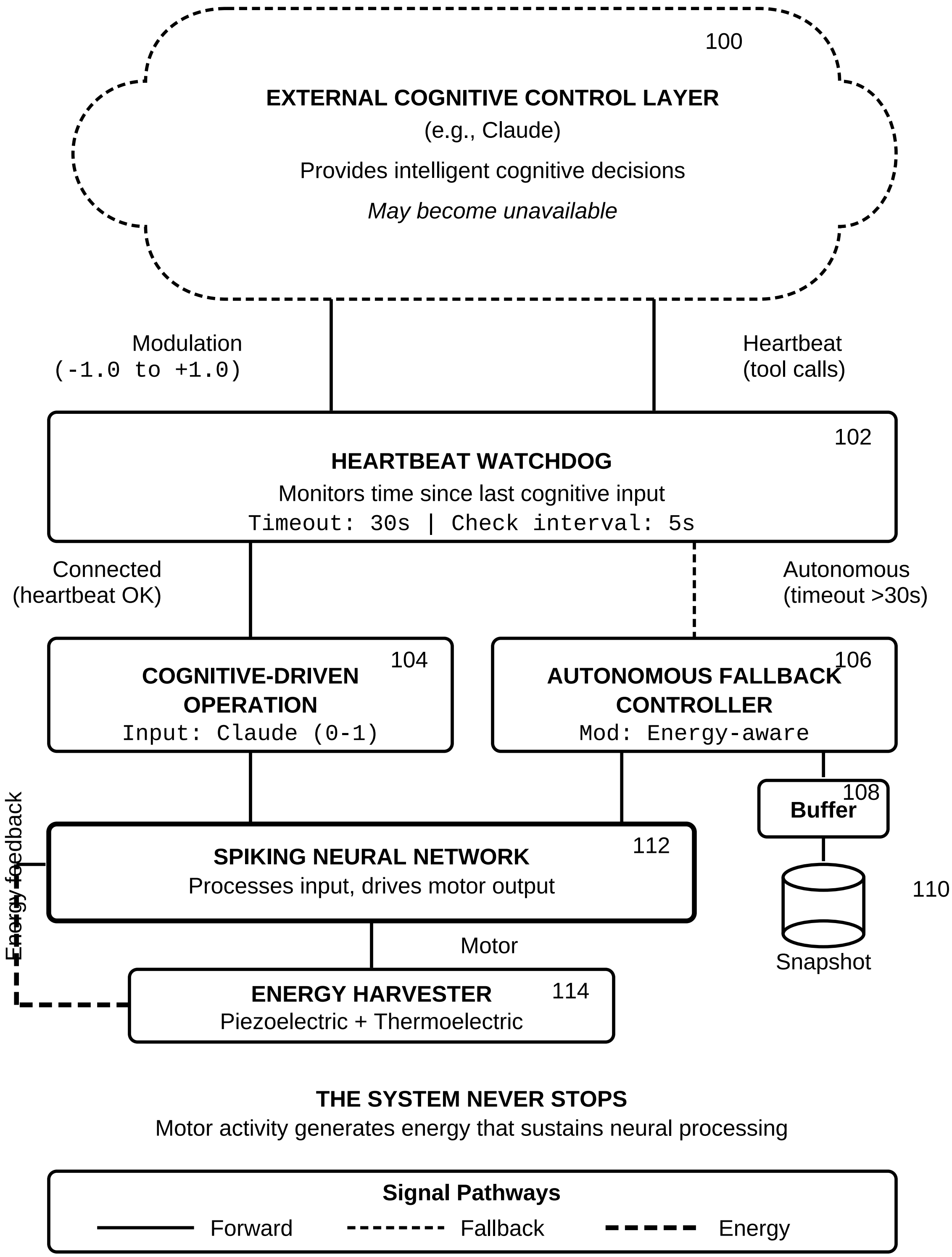


FIG. 1

STATE TRANSITION DIAGRAM

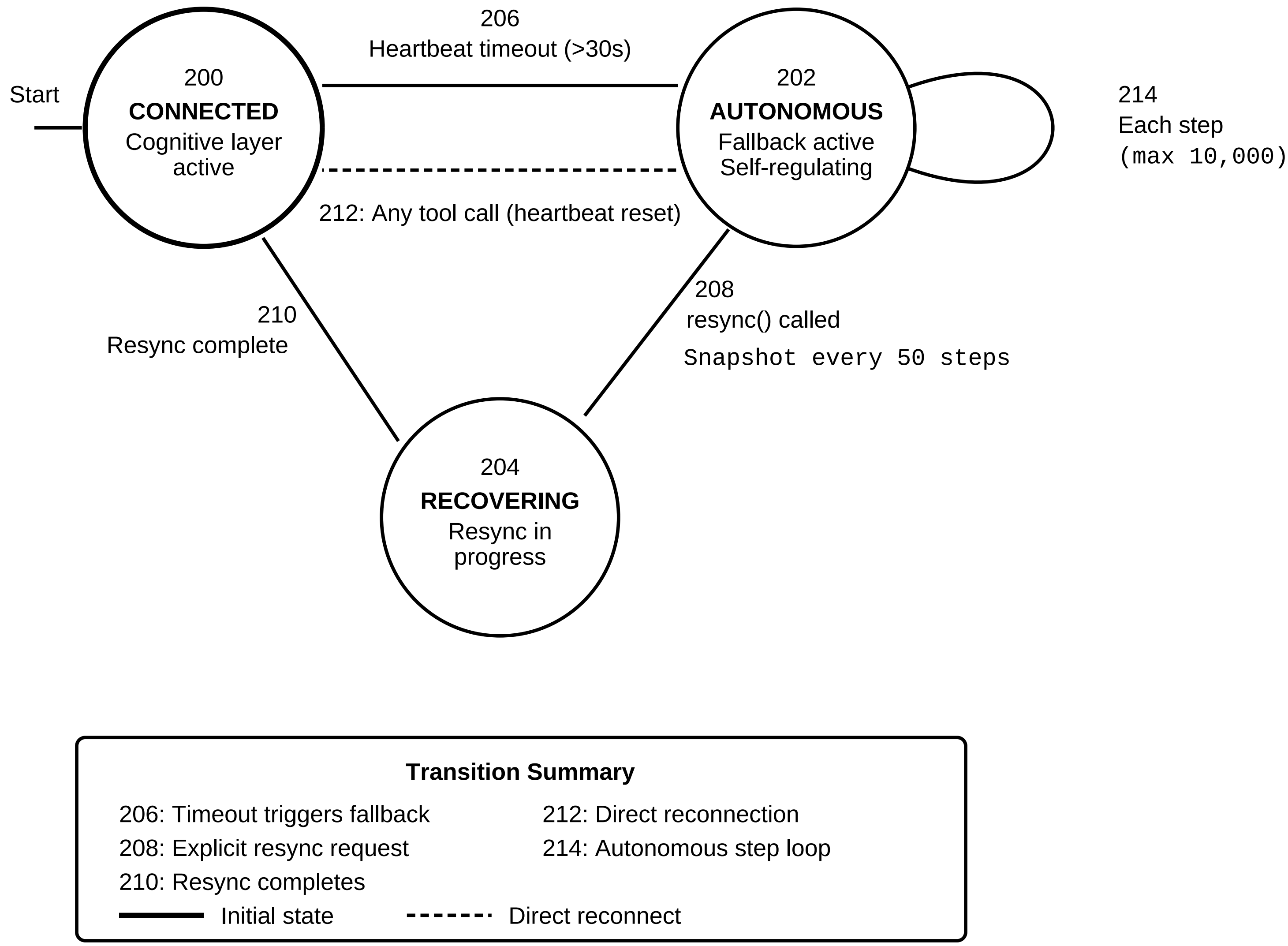
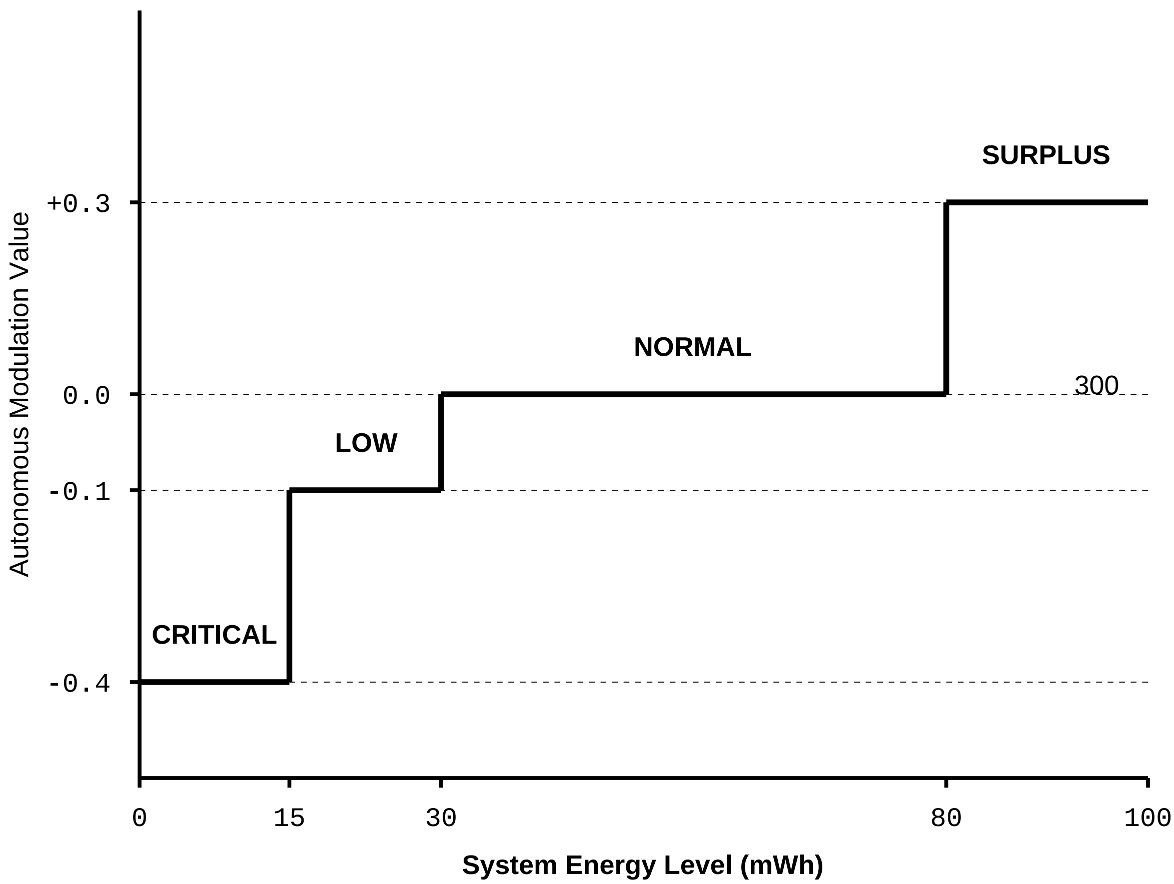


FIG. 2

ENERGY-AWARE MODULATION CURVE



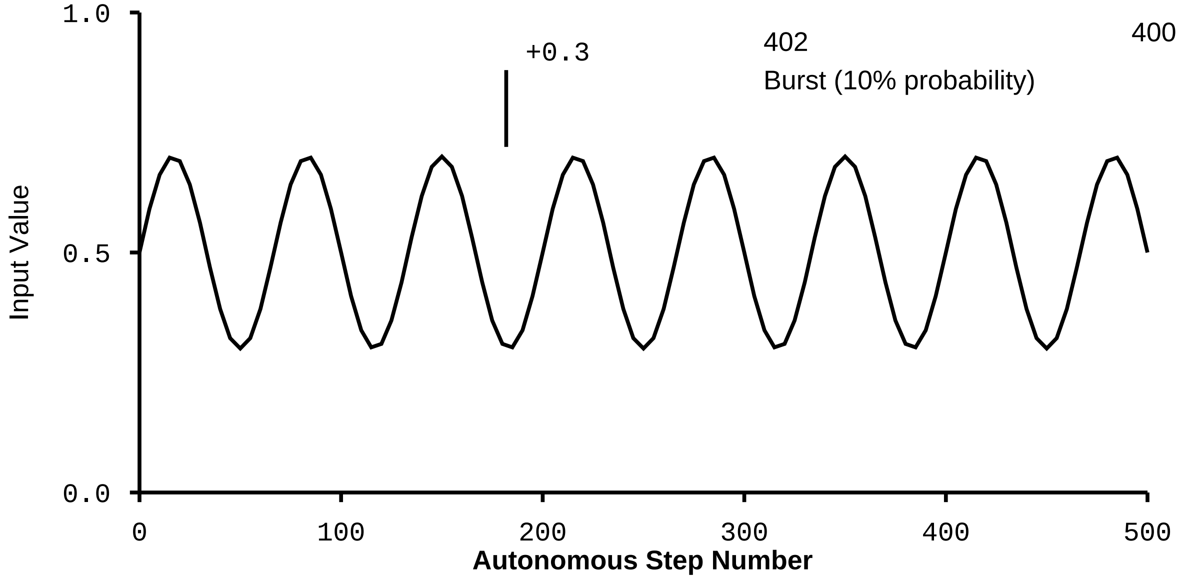
302

Behavioral Effects

< 15 mWh	mod = -0.4	Maximum conservation
15-30 mWh	mod = -0.1	Cautious operation
30-80 mWh	mod = 0.0	Standard processing
> 80 mWh	mod = +0.3	Exploration enabled

FIG. 3

AUTONOMOUS INPUT GENERATOR OUTPUT



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Autonomous Input Formula

circadian = 0.5 + 0.2 x sin(2pi x t x 3)

burst = 0.3 (with 10% probability per step)

Energy Recovery During Autonomous Operation

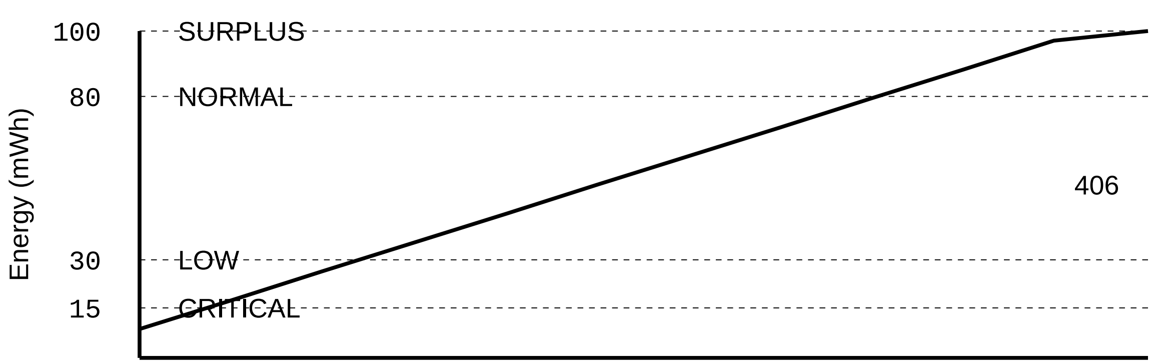


FIG. 4

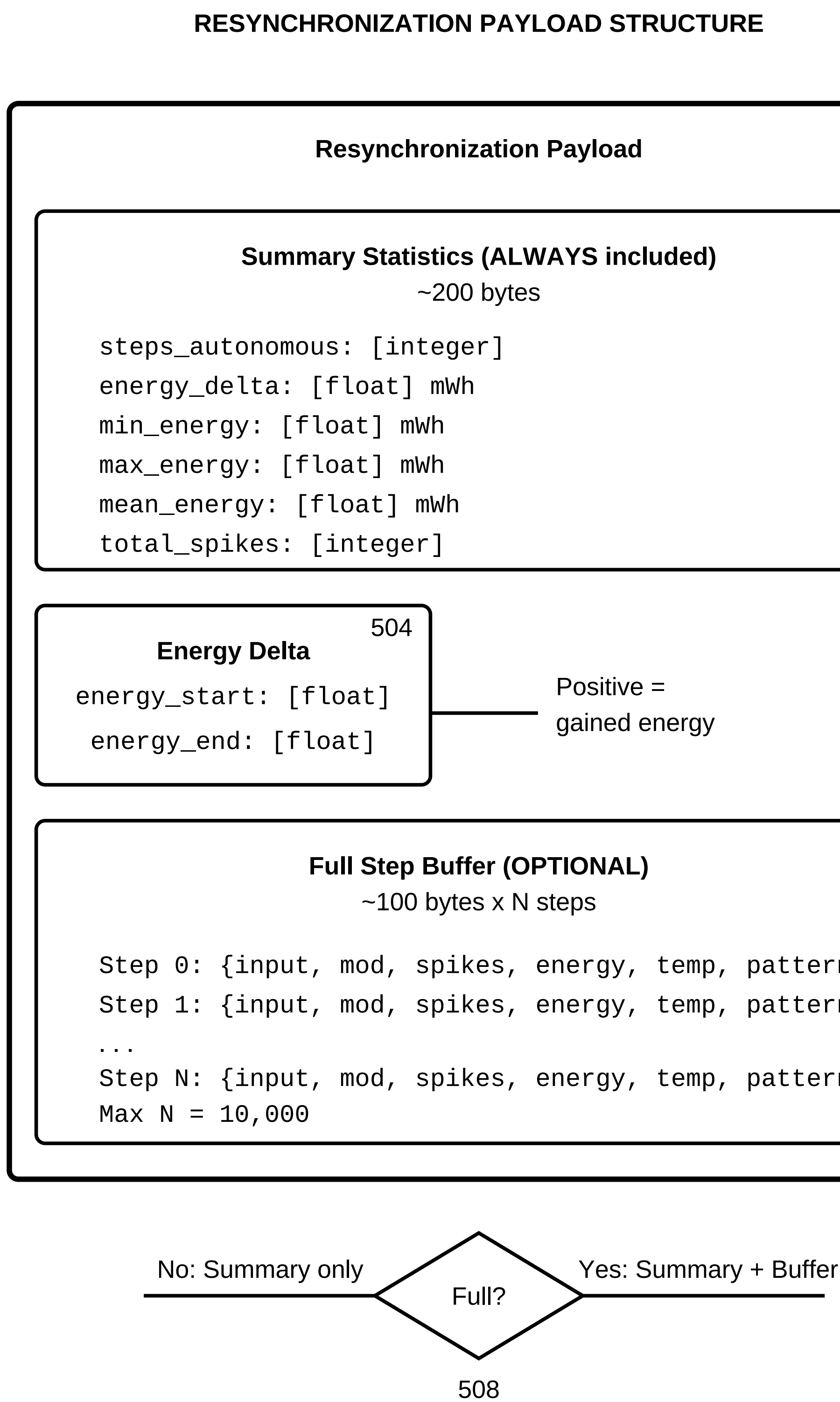
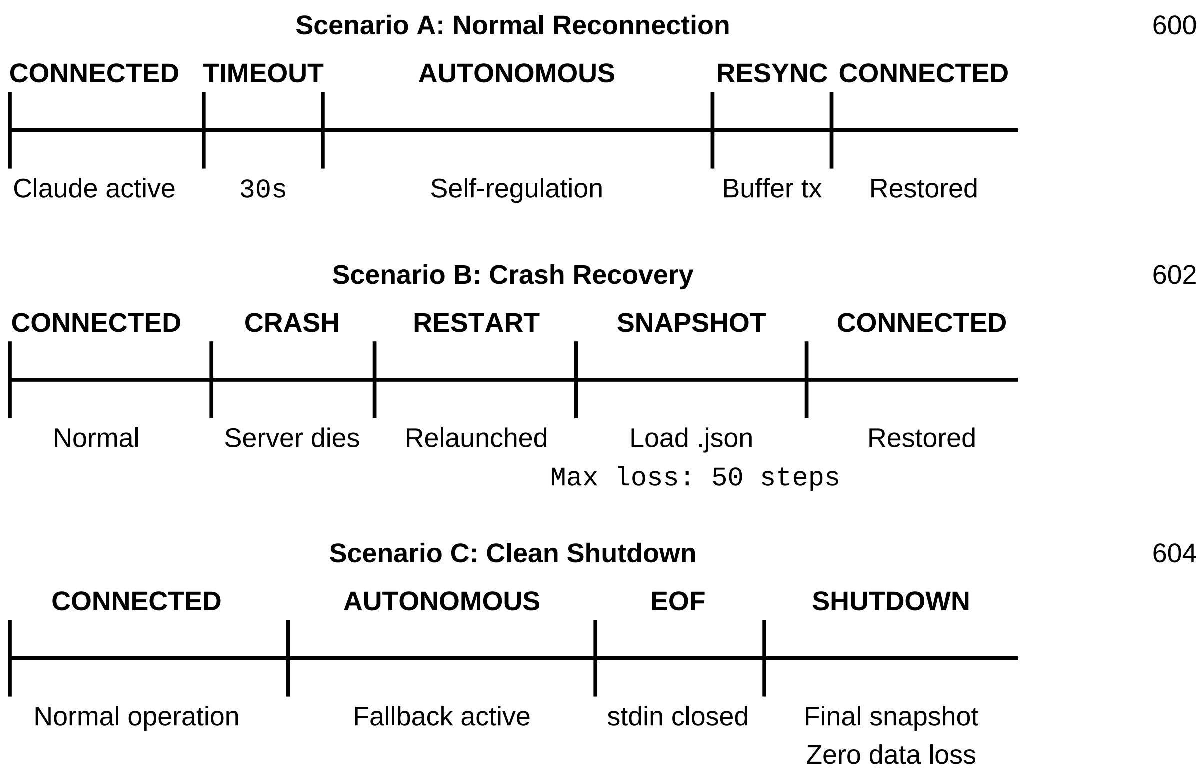

RESYNCHRONIZATION PAYLOAD STRUCTURE
500
Resynchronization Payload
502
Summary Statistics (ALWAYS included)
~200 bytes
steps_autonomous: [integer]
energy_delta: [float] mWh
min_energy: [float] mWh
max_energy: [float] mWh
mean_energy: [float] mWh
total_spikes: [integer]
504
Energy Delta
energy_start: [float]
energy_end: [float]
Positive = gained energy
506
Full Step Buffer (OPTIONAL)
~100 bytes x N steps
Step 0: {input, mod, spikes, energy, temp, pattern}
Step 1: {input, mod, spikes, energy, temp, pattern}
...
Step N: {input, mod, spikes, energy, temp, pattern}
Max N = 10,000
No: Summary only
Full?
Yes: Summary + Buffer
508

FIG. 5

RECOVERY SCENARIOS



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Recovery Guarantees

Normal reconnection: Full continuity, resync payload transmitted

Crash recovery: Max 50 steps lost (snapshot every 50 steps)

Clean shutdown: Zero data loss (final snapshot on EOF)

Max autonomous duration: 10,000 steps (~50 hours)

System actively processing

FIG. 6

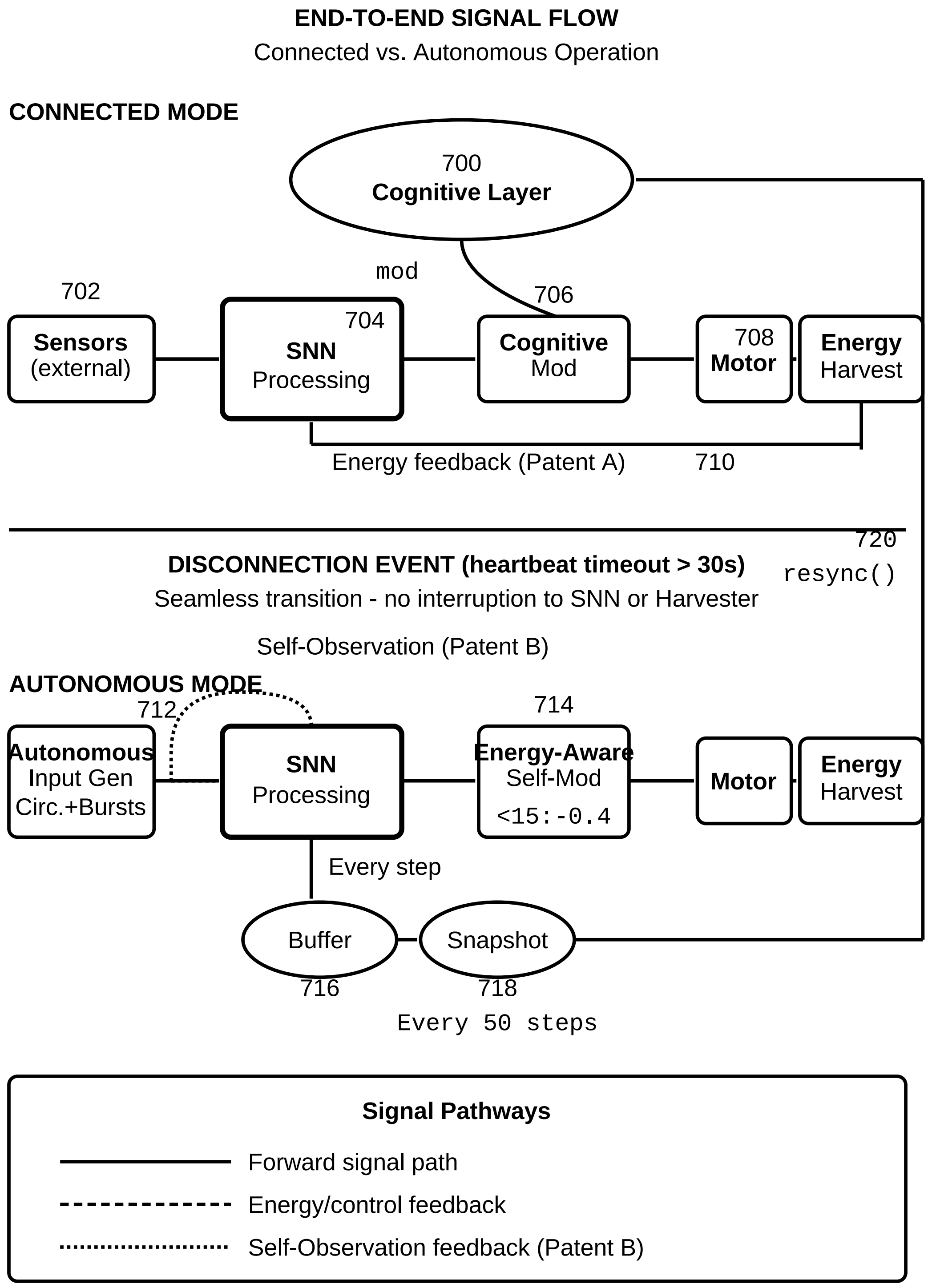
END-TO-END SIGNAL FLOW
Connected vs. Autonomous Operation
CONNECTED MODE
700
Cognitive Layer
mod
702
Sensors
(external)
704
SNN
Processing
706
Cognitive
Mod
708
Motor
Energy
Harvest
Energy feedback (Patent A)
710
720
resync()
DISCONNECTION EVENT (heartbeat timeout > 30s)
Seamless transition - no interruption to SNN or Harvester
Self-Observation (Patent B)
AUTONOMOUS MODE
712
Autonomous
Input Gen
Circ.+Bursts
SNN
Processing
714
Energy-Aware
Self-Mod
<15:-0.4
Motor
Energy
Harvest
Every step
Buffer
716
Snapshot
718
Every 50 steps
Signal Pathways
Forward signal path
Energy/control feedback
Self-Observation feedback (Patent B)

FIG. 7